



JERUSALEM COLLEGE OF ENGINEERING

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DEPARTMENT OF SCIENCE AND HUMANITES (MATHEMATICS DIVISION)

QUESTION BANK

Regulation	:	2023
Course Name	:	APPLIED LINEAR ALGEBRA
Course Code	:	JMA2402
Year / Semester / Branch	:	II/ IV/ CSE/IT/AI&ML/AI&DS/CS&BS/CS/ECE
Unit No	:	I
Unit Title	:	MATRICES AND SYSTEM OF LINEAR EQUATIONS
No. of Hours	:	12
Weightage in Exam (%)	:	20 %

Part A

S. No	Question	Marks	BTL	CO
1.	Define rank of a matrix. Let A be an $m \times n$ matrix. The rank of A is said to be r , if (i) there is at least one minor of A of order r which does not vanish & (ii) every minor of A of order $r + 1$ and higher vanishes	2	K1	CO1
2.	Define solution of a system of linear equations. A set of values of $x_1, x_2, \dots, x_n$ which satisfy all the m equations of a system of linear equations $AX = B$ simultaneously is called a solution of the system of linear equations.	2	K1	CO1
3.	Define consistent and inconsistent system of linear equations. If a system of linear equations $AX = B$ has a solution then it is said to be consistent, otherwise it is said to be inconsistent.	2	K1	CO1
4.	State Rouche's theorem. The system of linear equations $AX = B$ is consistent if and only if the coefficient matrix and augmented matrix are of the same rank (i.e.) $R[A] = R[A, B]$	2	K1	CO1
5.	What is the condition for a system of linear equations to have (i) exactly one solution (ii) infinitely many solutions. The system of linear equations $AX = B$ with n unknowns, is consistent with exactly one solution if $R[A] = R[A, B] = n$ and is consistent with infinitely many solutions if $R[A] = R[A, B] < n$	2	K1	CO1



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6.	Find a and b if the equations $2x - 3y = 5$ & $ax + by = -10$ have many solutions. Here $[A, B] = \begin{bmatrix} 2 & -3 & 5 \\ a & b & -10 \end{bmatrix}$ $\sim \begin{bmatrix} 2 & -3 & 5 \\ 0 & b + \frac{3a}{2} & -10 - \frac{5a}{2} \end{bmatrix} \quad R_2 \rightarrow R_2 - \frac{a}{2}R_1$ The system has many solutions if $R[A] = R[A, B] < 2$ (i.e.) $R[A] = R[A, B] = 1$ Hence $b + \frac{3a}{2} = 0$ & $-10 - \frac{5a}{2} = 0 \Rightarrow a = -4$ & $b = 6$	2	K2	CO1
7.	Find the condition satisfied by a, b, c so that the system of equations $x + 2y - 3z = a$, $3x - y + 2z = b$ & $x - 5y + 8z = c$ may have a solution. $[A, B] = \begin{bmatrix} 1 & 2 & -3 & a \\ 3 & -1 & 2 & b \\ 1 & -5 & 8 & c \end{bmatrix}$ $\sim \begin{bmatrix} 1 & 2 & -3 & a \\ 0 & -7 & 11 & b - 3a \\ 0 & -7 & 11 & c - a \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$ $\sim \begin{bmatrix} 1 & 2 & -3 & a \\ 0 & -7 & 11 & b - 3a \\ 0 & 0 & 0 & 2a - b + c \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$ The coefficient matrix A has rank 2. For the system to have a solution $[A, B]$ should also have rank 2, by Rouché's theorem (i.e.) $2a - b + c = 0$	2	K2	CO1
8.	Write the algorithm for Gauss-Jordan method: Step 1: Write the given system of equations in the matrix form (i.e.) $AX = B$ Step 2: Construct the augmented matrix (i.e.) $[A, B]$ Step 3: Using elementary row operations reduce the matrix A in $[A, B]$ to identity matrix. Step 4: Solve for the unknowns using back substitution method.	2	K1	CO1
9.	Write the diagonal dominance rule for a system of 3 linear equations with 3 unknowns. The system of equations $a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$ is said to be diagonally dominant if $a_1 > b_1 + c_1$, $b_2 > a_2 + c_2$ & $c_3 > a_3 + b_3$	2	K1	CO1
10.	What is the sufficient condition for convergence of Gauss-Seidel method? The sufficient condition for convergence of Gauss-Seidel method is that the system must be diagonally dominant.	2	K2	CO1
11.	Check whether the system $2x + 8y - z = 6$, $6x + y + 2z = 8$, $-3x + y + 7z = 5$ is diagonally dominant, if not convert it.	2	K2	CO1



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	First, we need to verify the conditions for diagonally dominance. $ a_1 \geq b_1 + c_1 $ $ b_2 \geq a_2 + c_2 $ $ c_3 \geq a_3 + b_3 $ $ 2 < 8 + -1 $ $\Rightarrow 1 < 6 + 2 $ $ 7 > -3 + 7 $ $\Rightarrow$ Equation (1) and (2) doesn't satisfies the condition of diagonally dominance. Therefore, Interchange the order of (1) and (2), Hence we have $6x + y + 2z = 8$ $2x + 8y - z = 6$ $-3x + y + 7z = 5$ which is the required diagonally dominant system.			
12.	Check whether the system $27x + 6y - z = 85$, $6x + 15y + 2z = 72$, $x + y + 54z = 110$ is diagonally dominant. First, we need to verify the conditions for diagonally dominance. $ a_1 \geq b_1 + c_1 $ $ b_2 \geq a_2 + c_2 $ $ c_3 \geq a_3 + b_3 $ $ 27 > 6 + -1 $ $\Rightarrow 15 > 6 + 2 $ $ 54 > 1 + 1 $ $\Rightarrow$ The given system satisfies the condition of diagonally dominance. Therefore, the given system is diagonally dominant.	2	K2	CO1
13.	Write the algorithm for Gauss-Seidel method: Step 1: Write the given system of equations in the diagonal dominant form, say $a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$ Step 2: The iterative formulas for x, y, z are given by $x^{(r+1)} = \frac{1}{a_1} [d_1 - b_1y^{(r)} - c_1z^{(r)}]$ $y^{(r+1)} = \frac{1}{b_2} [d_2 - a_2x^{(r+1)} - c_2z^{(r)}]$ $z^{(r+1)} = \frac{1}{c_3} [d_3 - a_3x^{(r+1)} - b_3y^{(r+1)}]$ Step 3: Use $y^{(0)} = z^{(0)} = 0$ to get first approximation. Use these values to get second approximation and continue (i.e.) put $r = 0, 1, 2, \dots$ Step 4: Continue above step until desired level of accuracy is reached in all three unknowns.	2	K2	CO1
14.	$10x + y + z = 13$ Write down the iterative formula for x, y, z of the system $x + 11y + z = 14$ $x + y + 12z = 15$ using Gauss-Seidal method.	2	K2	CO1



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	The given equations of the system are diagonally dominant, then the iterative formula for x, y, z are given by $x = \frac{1}{10}(13 - y - z)$ $y = \frac{1}{11}(14 - x - z)$ $z = \frac{1}{12}(15 - x - y)$			
15.	Find the first iterative values of x, y, z for the $\begin{aligned} 28x + 4y - z &= 32 \\ x + 3y + 10z &= 24 \\ 2x + 17y + 4z &= 35 \end{aligned}$ Gauss-seidal method. The given equations of the system are diagonally dominant, then the iterative formula for x, y, z are given by Let us assume $y^0 = z^0 = 0$ $x^1 = \frac{1}{28}(32 - 4y^0 + z^0) = \frac{1}{28}(32 - 0) = 1.1429$ $y^1 = \frac{1}{17}(35 - 2x^0 - 4z^0) = \frac{1}{17}(35 - 2.2858) = 1.9244$ $z^1 = \frac{1}{10}(24 - x^0 - 3y^0) = \frac{1}{10}(24 - 1.1429 - 5.7732) = 1.7084$ Therefore initial iterative values of x, y, z are $x^1 = 1.1429, \quad y^1 = 1.9244, \quad z^1 = 1.7084$	2	K2	CO1

Part B

S. No	Question	Marks	BTL	CO
1.	Test the consistency of the following system of equations and solve if consistent. (i) $\begin{aligned} 3x + y + z &= 8 \\ -x + y - 2z &= -5 \\ x + y + z &= 6 \\ -2x + 2y - 3z &= -7 \end{aligned}$ (ii) $\begin{aligned} 2x_1 + x_2 + 2x_3 + x_4 &= 6 \\ x_1 - x_2 + x_3 + 2x_4 &= 6 \\ 4x_1 + 3x_2 + 3x_3 - 3x_4 &= -1 \\ 2x_1 + 2x_2 - x_3 + x_4 &= 10 \end{aligned}$ (iii) $\begin{aligned} 2x + y + 5z + w &= 5 \\ x + y + 3z - 4w &= -1 \\ 3x + 6y - 2z + w &= 8 \\ 2x + 2y + 2z - 3w &= 2 \end{aligned}$	8	K4	CO1



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2.	Test the consistency of the following system of equations and solve if consistent. (i) $x - 4y - 3z + 16 = 0$ $2x + 7y + 12z = 48$ $4x - y + 6z = 16$ $5x - 5y + 3z = 0$ (ii) $x_1 + 2x_2 - x_3 - 5x_4 = 4$ $x_1 + 3x_2 - 2x_3 - 7x_4 = 5$ $2x_1 - x_2 + 3x_3 = 3$	8	K4	CO1
3.	Test the consistency of the following system of equations (i) $x - 2y - 3z = 2$ $3x - 2y = -1$ $-2y - 3z = 2$ $y + 2z = 1$ (ii) $x_1 + x_2 + x_3 = 1$ $2x_1 - 2x_2 + 3x_3 = 1$ $x_1 - x_2 + 2x_3 = 5$ $x_1 + x_2 + x_3 = 2$	8	K4	CO1
4.	(i) Find the values of k for which the equations $x + y + z = 1, x + 2y + 4z = k$ & $x + 4y + 10z = k^2$ may have a solution. For these values of k find the solution also. (ii) Find the values of λ for which the equations $x + 2y + z = 4, 2x - y - z = 3\lambda$ & $4x - 7y - 5z = \lambda^2$ may have a solution. For these values of λ find the solution also.	8	K4	CO1
5.	(i) Find the values of k for which the equations $kx + y + z = 1, x + ky + z = 1$ & $x + y + kz = 1$ have (a) infinitely many solutions (b) no solution (c) unique solution. (ii) Find the values of k for which the equations $kx - 2y + z = 1, x - 2ky + z = -2$ & $x - 2y + kz = 1$ have (a) no solution (b) many solutions (c) unique solution. (iii) For what values of λ, μ does the system of equations $x + y + 2z = 2, 2x - y + 3z = 2$ & $5x - y + \lambda z = \mu$ have (a) infinite number of solutions (b) no solution (c) unique solution. (iv) For what values of λ, μ does the system of equations $x + y + z = 6, x + 2y + 3z = 10$ & $x + 2y + \lambda z = \mu$ have (a) many solutions (b) no solution (c) only one solution.	8	K4	CO1
6.	Find the non-trivial solution of the following system of equations if it exists. (i) $x - y + 2z - 3w = 0$ $3x + 2y - 4z + w = 0$ $5x - 3y + 2z + 6w = 0$ $x - 9y + 14z - 2w = 0$	8	K4	CO1



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	(ii) $x + 2y + z + 2w = 0$ $x + 3y + 2z + 2w = 0$ $2x + 4y + 3z + 6w = 0$ $3x + 7y + 4z + 6w = 0$			
7.	Solve the following system of equations by Gauss Jordan method. (i) $x + y + z = 9$ $2x - 3y + 4z = 13$ $3x + 4y + 5z = 40$ (ii) $2x_1 + x_2 + x_3 = 10$ $3x_1 + 2x_2 + 3x_3 = 18$ $x_1 + 4x_2 + 9x_3 = 16$ (iii) $x_1 + 3x_2 + 3x_3 = 16$ $x_1 + 4x_2 + 3x_3 = 18$ $x_1 + 3x_2 + 4x_3 = 19$	8	K4	CO1
8.	Solve the following system of equations by Gauss Seidel method correct to 4 decimal places. (i) $9x_1 + 2x_2 + 4x_3 = 10$ $2x_1 - 4x_2 + 10x_3 = -15$ $x_1 + 10x_2 + 4x_3 = 6$ (ii) $x + 5y - z = 10$ $x + y + 8z = 20$ $4x + 2y + z = 14$ (iii) $x_1 + x_2 + 54x_3 = 1$ $27x_1 + 6x_2 - x_3 = 85$ $6x_1 + 15x_2 + 2x_3 = 72$	8	K4	CO1



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QUESTION BANK

Regulation	:	2023
Course Name	:	APPLIED LINEAR ALGEBRA
Course Code	:	JMA2402
Year / Semester / Branch	:	II/ IV/ CSE/IT/AI&ML/AI&DS/CS&BS/CS/ECE
Unit No	:	II
Unit Title	:	VECTOR SPACES
No. of Hours	:	12
Weightage in Exam (%)	:	20 %

Part A

S. No	Question	Marks	BTL	CO
1.	Define field with an example A field F is a set on which two operations addition and multiplication are defined such that the following conditions are satisfied. (i) Commutativity of addition and multiplication (ii) Associativity of addition and multiplication (iii) Existence of identity for addition and multiplication (iv) Existence of inverse for addition and multiplication (v) Distributivity of multiplication over addition Ex: The set of real numbers (i.e.) $\mathbb{R}$ is a field with addition and multiplication as operations.	2	K1	CO2
2.	Define vector space. A vector space V over a field F is a set on which two binary operations are defined such that the following conditions are satisfied. (i) $x + y = y + x \quad \forall x, y \in V$ (ii) $(x + y) + z = x + (y + z) \quad \forall x, y, z \in V$ (iii) There exists $0 \in V$ such that $x + 0 = x \quad \forall x \in V$ (iv) There exists $x' \in V$ such that $x + x' = 0 \quad \forall x \in V$ (v) $1x = x \quad \forall x \in V$ (vi) For $a, b \in F$ & $x \in V$, we have $(ab)x = a(bx)$ (vii) For $a \in F$ & $x, y \in V$, we have $a(x + y) = ax + ay$ (viii) For $a, b \in F$ & $x \in V$, we have $(a + b)x = ax + bx$	2	K1	CO2
3.	Define subspace. A subset W of a vector space V over a field F is called a subspace of V, if W is a vector space over F with the same operations defined on V.	2	K1	CO2



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4.	Give an example for vector space and its subspace. The set of all 3-tuples of real numbers $\mathbb{R}^3 = \{(x, y, z)/x, y, z \in \mathbb{R}\}$ is a vector space over the field of real numbers $\mathbb{R}$ with component wise addition and scalar multiplication as operations. The set $W = \{(x, 2y, 3z)/x, y, z \in \mathbb{R}\}$ is a proper subset of $\mathbb{R}^3$ and is also a vector space over $\mathbb{R}$ under the same operations. Thus W is a subspace of $\mathbb{R}^3$.	2	K1	CO2
5.	State the necessary and sufficient conditions for a non-empty subset W of a vector space V defined over a field F to be a subspace of V. The necessary and sufficient conditions for a non-empty subset W of a vector space V defined over a field F to be a subspace of V are (i) $0 \in W$ (ii) $a \in F \text{ \& } x, y \in W \Rightarrow ax + y \in W$	2	K1	CO2
6.	Is $W = \{(a_1, a_2, a_3) \in \mathbb{R}^3 / a_1 + 2a_2 - 3a_3 = 1\}$ a subspace of vector space $\mathbb{R}^3$ defined over the field of real numbers $\mathbb{R}$ under coordinate wise addition and scalar multiplication? To check if $0 \in W$. For $(0, 0, 0) \in \mathbb{R}^3$, $0 + 2(0) - 3(0) = 0 \neq 1$. Hence $0 \notin W$. Thus W is not a subspace of vector space $\mathbb{R}^3$ defined over the field of real numbers $\mathbb{R}$.	2	K2	CO2
7.	Define linear dependence and independence of vectors. Let V be a vector space over field F. A finite set of vectors $\{u_1, u_2, \dots, u_n\}$ in V is said to be linearly dependent if one vector can be expressed as linear combination of other vectors, if not it is linearly independent.	2	K1	CO2
8.	Define a basis for a vector space V defined over field F. A subset S of a vector space V defined over field F is said to be a basis if (i) S consists of linearly independent vectors & (ii) S generates V (i.e.) every element of V is a linear combination of elements in S.	2	K1	CO2
9.	Write the standard basis for $\mathbb{R}^2$ and $\mathbb{R}^3$. Standard basis for $\mathbb{R}^2$ is $\{(1, 0), (0, 1)\}$ Standard basis for $\mathbb{R}^3$ is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$	2	K2	CO2
10.	Write the standard basis for $M_{2 \times 2}(\mathbb{R})$ and $M_{2 \times 3}(\mathbb{R})$. Standard basis for $M_{2 \times 2}(\mathbb{R})$ is $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ Standard basis for $M_{2 \times 3}(\mathbb{R})$ is $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \right\}$	2	K2	CO2
11.	Write the standard basis for $P_2(\mathbb{R})$, and $P_3(\mathbb{R})$. Standard basis for $P_2(\mathbb{R})$ is $\{1, x, x^2\}$ Standard basis for $P_3(\mathbb{R})$ is $\{1, x, x^2, x^3\}$	2	K2	CO2



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12.	Define dimension of a Vector Space V over a Field $\mathbb{F}$. Let V be a vector space over field $\mathbb{F}$. If V has a basis with n vectors in it, then n is called the dimension of V. (i.e) $\dim_{\mathbb{F}}(V) = n$	2	K1	CO2
13.	Find the dimension of the vector space $\mathbb{R}^3$ over $\mathbb{R}$. W.K.T the Standard basis for $\mathbb{R}^3$ is $\{(1,0,0), (0,1,0), (0,0,1)\}$ And the number of vectors in the Basis of $\mathbb{R}^3$ is 3 Hence the Dimension of vector space $\mathbb{R}^3$ over field $\mathbb{R}$ is 3 (i.e.) $\dim_{\mathbb{R}}(\mathbb{R}^3) = 3$	2	K2	CO2
14.	Find the dimension of the vector space $M_{2 \times 3}(\mathbb{R})$. W.K.T the Standard basis for $M_{2 \times 3}(\mathbb{R})$ is $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$ And the number of vectors in the Basis of $M_{2 \times 3}(\mathbb{R})$ is 6 Hence the Dimension of vector space $M_{2 \times 3}(\mathbb{R})$ is 6 (i.e.) $\dim(M_{2 \times 3}(\mathbb{R})) = 6$	2	K2	CO2
15.	Find the dimension of the vector space $P_3(\mathbb{R})$. W.K.T the Standard basis for $P_3(\mathbb{R})$ is $\{1, x, x^2, x^3\}$ And the number of vectors in the Basis of $\mathbb{R}^3$ is 4 Hence the Dimension of vector space $P_3(\mathbb{R})$ is 4 (i.e.) $\dim(P_3(\mathbb{R})) = 4$	2	K2	CO2

Part B

S. No.	Question	Marks	BTL	CO
1.	(i) Let $V = \{(a_1, a_2, \dots, a_n) / a_i \in \mathbb{R}, i = 1, 2, \dots, n\}$. Is V a vector space over $\mathbb{R}$ with the operations coordinate wise addition and scalar multiplication? (OR) Show that the set of n –tuples of real numbers $\mathbb{R}^n$ is a vector space over the field of real numbers $\mathbb{R}$ with coordinate wise addition and scalar multiplication as operations. (ii) Let V be the set of all $m \times n$ matrices with real numbers as elements. Is V a vector space over $\mathbb{R}$ with the usual matrix addition and scalar multiplication as operations?	16	K4	CO2



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	(iii) Show that the set of all real valued polynomials of degree less than or equal to n denoted by $P_n(\mathbb{R})$ is a vector space with operations of addition & scalar multiplication. (iv) Let $V = \{(a_1, a_2)/a_i \in \mathbb{R}, i = 1, 2\}$. If $(a_1, a_2), (b_1, b_2) \in V$ & $c \in \mathbb{R}$ define $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$ & $c(a_1, a_2) = (ca_1, ca_2)$. Is V a vector space over $\mathbb{R}$ with the defined operations? (v) Let $V = \{(a_1, a_2)/a_i \in \mathbb{R}, i = 1, 2\}$. Define addition of elements of V coordinate wise and for $(a_1, a_2) \in V$ & $c \in \mathbb{R}$, define $c(a_1, a_2) = \begin{cases} (0, 0), & \text{if } c = 0 \\ (ca_1, \frac{a_2}{c}), & \text{if } c \neq 0. \end{cases}$ Is V a vector space over $\mathbb{R}$ with the defined operations?			
2.	(i) Prove that $W = \{(a_1, a_2, a_3) \in \mathbb{R}^3 / a_1 = 3a_2 \text{ \& } a_3 = -a_2\}$ is a subspace of the vector space $\mathbb{R}^3$ defined over the field of real numbers $\mathbb{R}$ under coordinate wise addition and scalar multiplication. (ii) Prove that $W = \{A \in M_{2 \times 2}(\mathbb{R}) / A^T = A\}$ is a subspace of the vector space $M_{2 \times 2}(\mathbb{R})$ defined over the field of real numbers $\mathbb{R}$ under matrix addition and scalar multiplication. (iii) Prove that $W = \{a + bx + cx^2 \in P_2(\mathbb{R}) / c = a + b\}$ is a subspace of vector space $P_2(\mathbb{R})$ defined over the field of real numbers $\mathbb{R}$ under addition and scalar multiplication.	8	K4	CO2
3.	Check whether the following set of vectors are linearly dependent, if so express any one vector as a linear combination of other vectors (i) $\{(-2, 2, 2), (1, 2, -1), (-3, -3, 3)\}$ (ii) $\{(1, 1, 2, 4), (2, -1, -5, 2), (1, -1, -4, 0), (2, 1, 1, 6)\}$ (iii) $\{(3, 4, 1), (1, -2, 1), (-2, -1, 1)\}$, (iv) $\{-2x^3 - 11x^2 + 3x + 2, x^3 - 2x^2 + 3x - 1, 2x^3 + x^2 + 3x - 2\}$ (v) $\left\{ \begin{bmatrix} 8 & 1 \\ -5 & 6 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \right\}$	8	K4	CO2
4.	Show that the following set of vectors (i) $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ generate $\mathbb{R}^3$ (ii) $\left\{ \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ generate $M_{2 \times 2}(\mathbb{R})$ (iii) $\{x^2 + 3x - 2, 2x^2 + 5x - 3, -x^2 - 4x + 4\}$ generate $P_2(\mathbb{R})$	8	K4	CO2
5.	Verify whether the following set of vectors are linearly independent, if so show that they generate the entire vector space (or) Determine which of the following sets form basis. (i) $\{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$ for $\mathbb{R}^3$ (ii) $\{1 - 2x - 2x^2, -2 + 3x - x^2, 1 - x + 6x^2\}$ for $P_2(\mathbb{R})$ (iii) $\{x^3 + 2x^2, -x^2 + 3x + 1, x^3 - x^2 + 2x - 1\}$ for $P_3(\mathbb{R})$	8	K4	CO2



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	(iv) $\left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ for $M_{2 \times 2}(\mathbb{R})$ (v) $\left\{ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 3 & -2 \\ 1 & 2 \end{bmatrix} \right\}$ for $\mathbb{R}^3$			
6.	Determine whether the given vector v_1 is in the span of S (i) $v_1 = (2, -1, 1), S = \{(1, 0, 2), (-1, 1, 1)\}$ (ii) $v_1 = (-1, 1, 1, 2), S = \{(1, 0, 1, -1), (0, 1, 1, 1)\}$ (iii) $v_1 = 2x^3 - x^2 + x + 3, S = \{x^3 + x^2 + x + 1, x^2 + x + 1, x + 1\}$ (iv) $v_1 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}, S = \left\{ \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\}$ (v) $v_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, S = \left\{ \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\}$	8	K4	CO2



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DEPARTMENT OF SCIENCE AND HUAMITIES (MATHEMATIS DIVISION)

QUESTION BANK

Regulation	:	2023
Course Name	:	APPLIED LINEAR ALGEBRA
Course Code	:	JMA2402
Year / Semester / Branch	:	II/ IV/ CSE/IT/AI&ML/AI&DS/CS&BS/CS/ECE
Unit No	:	III
Unit Title	:	LINEAR TRANSFORMATION
No. of Hours	:	12
Weightage in Exam (%)	:	20 %

Part A

S. No	Question	Marks	BTL	CO
1.	Define linear transformation. Let V and W be vector spaces over the field F . A function $T: V \rightarrow W$ is called a linear transformation if $\forall x, y \in V$ & $c \in F$ (i) $T(x + y) = T(x) + T(y)$ (ii) $T(cx) = cT(x)$	2	K1	CO3
2.	If $T: V \rightarrow W$ is a linear transformation then show that $T(0) = 0$. Let $x \in V \Rightarrow T(x) \in W$ Consider $T(x) + 0 = T(x)$ [Since $T(x) \in W$ & W is a vector space] $= T(x + 0)$ [Since $x \in V$ & V is a vector space] $= T(x) + T(0)$ [Since T is a linear transformation] $\Rightarrow T(x) + 0 = T(x) + T(0)$ $\Rightarrow T(0) = 0$ [Cancelling $T(x)$ on either side]	2	K2	CO3
3.	If $T: V \rightarrow W$ is a linear transformation then show that $T(x - y) = T(x) - T(y)$. Let $x, y \in V \Rightarrow T(x), T(y) \in W$ Consider $T(x - y) = T(x + (-y))$ $= T(x) + T(-y)$ [Since T is a linear transformation] $= T(x) + T((-1)y)$ $= T(x) + (-1)T(y)$ [Since T is a linear transformation] $= T(x) - T(y)$ Thus $T(x - y) = T(x) - T(y)$	2	K2	CO3
4.	Prove that T is a linear transformation if $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by $T(x, y) = (0, y)$.	2	K2	CO3



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	Let $\forall u = (x_1, y_1), v = (x_2, y_2) \in \mathbb{R}^2$ & $\forall \alpha \in \mathbb{R}$ Now $T(\alpha u + v) = T(\alpha(x_1, y_1) + (x_2, y_2))$ $= T((\alpha x_1, \alpha y_1) + (x_2, y_2))$ $= T(\alpha x_1 + x_2, \alpha y_1 + y_2)$ $= (0, \alpha y_1 + y_2) \dots\dots\dots(1)$ Now $\alpha T(u) + T(v) = \alpha T(x_1, y_1) + T(x_2, y_2)$ $= \alpha(0, y_1) + (0, y_2)$ $= (0, \alpha y_1) + (0, y_2)$ $= (0, \alpha y_1 + y_2) \dots\dots\dots(2)$ Thus from (1) & (2), we have $T(\alpha u + v) = \alpha T(u) + T(v)$ Hence T is Linear Transformation			
5.	Prove that T is a linear transformation if $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by $T(x, y) = (x, -y)$. Let $\forall u = (x_1, y_1), v = (x_2, y_2) \in \mathbb{R}^2$ & $\forall \alpha \in \mathbb{R}$ Now $T(\alpha u + v) = T(\alpha(x_1, y_1) + (x_2, y_2))$ $= T((\alpha x_1, \alpha y_1) + (x_2, y_2))$ $= T(\alpha x_1 + x_2, \alpha y_1 + y_2)$ $= (\alpha x_1 + x_2, -(\alpha y_1 + y_2)) \dots\dots\dots(1)$ Now $\alpha T(u) + T(v) = \alpha T(x_1, y_1) + T(x_2, y_2)$ $= \alpha(x_1, -y_1) + (x_2, -y_2)$ $= (\alpha x_1, -\alpha y_1) + (x_2, -y_2)$ $= (\alpha x_1 + x_2, -(\alpha y_1 + y_2)) \dots\dots\dots(2)$ Thus from (1) & (2), we have $T(\alpha u + v) = \alpha T(u) + T(v)$ Hence T is Linear Transformation.	2	K2	CO3
6.	Define range and null space of a linear transformation. Let V and W be vector spaces over the field F and $T: V \rightarrow W$ be a linear transformation. (i) Range of T is defined as the set of all images of vectors in V (i.e.) $R(T) = \{T(x)/x \in V\}$ (ii) Null space of T is the set of all vectors in V whose image is 0 (i.e.) $N(T) = \{x \in V/T(x) = 0\}$	2	K1	CO3
7.	Define rank and nullity of a linear transformation. Let V and W be vector spaces over the field F and $T: V \rightarrow W$ be a linear transformation. If $R(T)$ & $N(T)$ are finite dimensional, then (i) Rank of T is defined as $rank(T) = \dim(R(T))$ (ii) Nullity of T is defined as $nullity(T) = \dim(N(T))$	2	K1	CO3



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8.	Let V and W be two Vector spaces over a field $\mathbb{F}$, and a Linear Transformation $Z: V \rightarrow W$ defined by $Z(v) = 0, \forall v \in V$ then Determine the Subspaces $N(Z)$ and $R(Z)$. $\text{Let } N(Z) = \{v \in V : Z(v) = 0, \forall v \in V\}$ $= V$ $R(Z) = \{Z(v) \in W : Z(v) = 0, \forall v \in V\}$ $= \{0\} \subset W$	2	K2	CO3
9.	Let V be a Vector space over a field $\mathbb{F}$, and a Linear Transformation $I: V \rightarrow V$ defined by $I(v) = v, \forall v \in V$ then Determine the Subspaces $N(I)$ and $R(I)$. $\text{Let } N(I) = \{v \in V : I(v) = v, \forall v \in V\}$ $= \{0\}$ $R(I) = \{I(v) \in W : I(v) = v, \forall v \in V\}$ $= V$	2	K2	CO3
10.	Find Null space of Linear Transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T(a, b, c) = (a - b, 2c)$. $\text{Let } N(T) = \{(a, b, c) \in \mathbb{R}^3 : T(a, b, c) = (0, 0)\}$ $= \{(a, b, c) \in \mathbb{R}^3 : (a - b, 2c) = (0, 0)\}$ (Since by Defn. of T) $= \{(a, b, c) \in \mathbb{R}^3 : a - b = 0, 2c = 0\}$ $= \{(a, b, c) \in \mathbb{R}^3 : a = b, c = 0\}$ $= \{(a, a, 0) \in \mathbb{R}^3\}$ $= \{a(1, 1, 0) \in \mathbb{R}^3 : \text{where } a \in \mathbb{R}\}$	2	K2	CO3
11.	State dimension theorem. Let V and W be vector spaces over the field F and $T: V \rightarrow W$ be a linear transformation. If V is finite dimensional, then $\text{rank}(T) + \text{nullity}(T) = \dim(V)$.	2	K1	CO3
12.	Define a one-one linear transformation. Let V and W be vector spaces over the field F and $T: V \rightarrow W$ be a linear transformation. If for $x, y \in V, T(x) = T(y) \Rightarrow x = y$ then T is said to be a one-one linear transformation.	2	K1	CO3
13.	Define an onto linear transformation. Let V and W be vector spaces over the field F and $T: V \rightarrow W$ be a linear transformation. T is said to be an onto linear transformation if $\forall y \in W$ there exists $x \in V$ such that $T(x) = y$ (or) if $\dim(R(T)) = \text{Rank of } T = \dim(W)$	2	K1	CO3
14.	Is $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1 + a_2, a_1)$ one-one? Let $x, y \in \mathbb{R}^2$ such that $T(x) = T(y) \Rightarrow T(x_1, x_2) = T(y_1, y_2)$ $\Rightarrow (x_1 + x_2, x_1) = (y_1 + y_2, y_1)$ $\Rightarrow x_1 + x_2 = y_1 + y_2 \text{ \& } x_1 = y_1$ $\Rightarrow x_1 = y_1 \text{ \& } x_2 = y_2$ $\Rightarrow (x_1, x_2) = (y_1, y_2)$ $\Rightarrow x = y$ Thus T is a one-one transformation.	2	K2	CO3



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15.	Is $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a_1, a_2) = (a_1 + a_2, a_1 - a_2, a_2)$ one-one? Let $x, y \in \mathbb{R}^2$ such that $T(x) = T(y)$ $\Rightarrow T(x_1, x_2) = T(y_1, y_2)$ $\Rightarrow (x_1 + x_2, x_1 - x_2, x_2) = (y_1 + y_2, y_1 - y_2, y_2)$ $\Rightarrow x_1 + x_2 = y_1 + y_2, \quad x_1 - x_2 = y_1 - y_2 \text{ \& } x_2 = y_2$ $\Rightarrow x_1 = y_1 \text{ \& } x_2 = y_2$ $\Rightarrow (x_1, x_2) = (y_1, y_2)$ $\Rightarrow x = y$ Thus T is a one-one transformation.	2	K2	CO3
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Part B

S. No	Question	Marks	BTL	CO
1.	(i) Let V and W be two vector spaces over a field F. Define zero transformation and identity transformation and prove that they are linear. (ii) Define reflection about x -axis and projection on x -axis and show that they are linear transformations. (iii) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined by $T(a_1, a_2) = (2a_1 + a_2, a_1)$, then prove that T is a linear transformation. (iv) Check if $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as given below, are linear transformations or not. (a) $T(a_1, a_2) = (a_1, a_2^2)$ (b) $T(a_1, a_2) = (a_1 + 1, a_2)$ (v) If $T: M_{m \times n}(\mathbb{R}) \rightarrow M_{n \times m}(\mathbb{R})$ is defined as $T(A) = A^t$, where A^t is the transpose of A, then prove that T is a linear transformation. (vi) If $T: P_n(\mathbb{R}) \rightarrow P_{n-1}(\mathbb{R})$ is defined as $T(f(x)) = f'(x)$, where $f'(x)$ denotes the derivative of $f(x)$, then show that T is linear. (vii) Let $V = C(\mathbb{R})$, the vector space of continuous real-valued functions on $\mathbb{R}$. Let $a, b \in \mathbb{R}, a < b$. Define $T: V \rightarrow \mathbb{R}$ by $T(f) = \int_a^b f(t) dt$ for all $f \in V$, then prove that T is a linear transformation.	8	K4	CO3
2.	(i) If $T: V \rightarrow W$ is a linear transformation, where V and W are vector spaces, then prove that T is one-one if and only if $N(T) = \{0\}$. (ii) Let V and W be vector spaces and $T: V \rightarrow W$ be a linear transformation, then prove that $N(T)$ and $R(T)$ are subspaces of V and W respectively.	8	K4	CO3
3.	(i) Prove that $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ is a linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify dimension theorem. Also check if T is one-one or onto.	16	K4	CO3



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(ii) Prove that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a_1, a_2) = (a_1 + a_2, 0, 2a_1 - a_2)$ is a linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify dimension theorem. Also check if T is one-one or onto. (iii) Prove that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a, b) = (a + b, a - b, b)$ is a linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify dimension theorem. Also check if T is one-one or onto. (iv) Prove that T defined as below is a linear Transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify RND theorem. Also check if T is one-to-one, onto. Define $T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ by $T(f(x)) = xf(x) + f'(x)$ (v) Prove that $T: M_{2 \times 3}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ defined by $T \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = \begin{pmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{pmatrix}$ is a linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify dimension theorem. Also check if T is one-one or onto.			
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DEPARTMENT OF SCIENCE AND HUAMITIES (MATHEMATIS DIVISION)

QUESTION BANK

Regulation	:	2023
Course Name	:	APPLIED LINEAR ALGEBRA
Course Code	:	JMA2402
Year / Semester / Branch		II/ IV/ CSE/IT/AI&ML/AI&DS/CS&BS/CS/ECE
Unit No	:	IV
Unit Title	:	DIAGONALIZATION
No. of Hours	:	12
Weightage in Exam (%)		20 %

Part A

S. No	Question	Marks	BTL	CO
1.	Define ordered basis. Let V be a finite dimensional vector space. An ordered basis for V is a finite sequence of linearly independent vectors of V that generates V .	2	K1	CO4
2.	Define matrix of a linear transformation. Let V be an n -dimensional vector space and W be an m -dimensional vector space over the field F . Let $\alpha = \{v_1, v_2, \dots, v_n\}$ and $\beta = \{w_1, w_2, \dots, w_m\}$ be ordered basis for V and W respectively. The matrix of the linear transformation $T: V \rightarrow W$ is an $m \times n$ matrix whose j^{th} column consists of the coefficients of $T(v_j)$ written as a linear combination of $\{w_1, w_2, \dots, w_m\}$ for all $j = 1, 2, \dots, n$ represented by $[T]_{\alpha}^{\beta}$	2	K2	CO4
3.	If T is a linear transformation of $\mathbb{R}^2$ defined by $T(a, b) = (a, 0)$, write the matrix of T relative to the standard basis of $\mathbb{R}^2$. Here the linear transformation is given by $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ defined as } T(a, b) = (a, 0).$ The standard basis of vector space $\mathbb{R}^2$ is given by $\alpha = \{(1,0), (0,1)\}$. Thus $v_1 = (1,0)$ & $v_2 = (0,1)$ $T(v_1) = T(1,0) = (1,0)$ [Using given definition of T] $= 1(1,0) + 0(0,1)$ $= 1v_1 + 0v_2$ Thus $T(v_1) = 1v_1 + 0v_2$ ---(1)	2	K2	CO4



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	$T(v_2) = T(0,1) = (0,0) \quad [\text{Using given definition of } T]$ $= 0(1,0) + 0(0,1)$ $= 0v_1 + 0v_2$ Thus $T(v_2) = 0v_1 + 0v_2 \quad \text{---(2)}$ From (1) and (2), the required matrix of linear transformation T relative to α is given by $[T]_\alpha = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$			
4.	Write the matrix of the linear transformation T of $\mathbb{R}^2$ defined by $T(a, b) = (0, b)$, relative to the standard basis of $\mathbb{R}^2$. Here the linear transformation is given by $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T(a, b) = (0, b)$. The standard basis of the vector space $\mathbb{R}^2$ is given by $\alpha = \{(1,0), (0,1)\}$. Thus $v_1 = (1,0)$ & $v_2 = (0,1)$ $T(v_1) = T(1,0) = (0,0) \quad [\text{Using given definition of } T]$ $= 0(1,0) + 0(0,1)$ $= 0v_1 + 0v_2$ Thus $T(v_1) = 0v_1 + 0v_2 \quad \text{---(1)}$ $T(v_2) = T(0,1) = (0,1) \quad [\text{Using given definition of } T]$ $= 0(1,0) + 1(0,1)$ $= 0v_1 + 1v_2$ Thus $T(v_2) = 0v_1 + 1v_2 \quad \text{---(2)}$ From (1) and (2), the required matrix of linear transformation T relative to α is given by $[T]_\alpha = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$	2	K2	CO4
5.	Write the matrix of the linear transformation T of $\mathbb{R}^2$ defined by $T(a, b) = (a, b)$, relative to the standard basis of $\mathbb{R}^2$. Here the linear transformation is given by $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T(a, b) = (a, b)$. The standard basis of the vector space $\mathbb{R}^2$ is given by $\alpha = \{(1,0), (0,1)\}$. Thus $v_1 = (1,0)$ & $v_2 = (0,1)$ $T(v_1) = T(1,0) = (1,0) \quad [\text{Using given definition of } T]$ $= 1(1,0) + 0(0,1)$ $= 1v_1 + 0v_2$ Thus $T(v_1) = 1v_1 + 0v_2 \quad \text{---(1)}$ $T(v_2) = T(0,1) = (0,1) \quad [\text{Using given definition of } T]$ $= 0(1,0) + 1(0,1)$ $= 0v_1 + 1v_2$ Thus $T(v_2) = 0v_1 + 1v_2 \quad \text{---(2)}$ From (1) and (2), the required matrix of linear transformation T relative to α is given by $[T]_\alpha = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	2	K2	CO4



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6.	If T is a linear transformation of $\mathbb{R}^2$ defined by $T(a, b) = (2a + b, a)$, write the matrix of T relative to the standard basis of $\mathbb{R}^2$. Here the linear transformation is given by $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T(a, b) = (2a + b, a)$. The standard basis of the vector space $\mathbb{R}^2$ is given by $\alpha = \{(1,0), (0,1)\}$. Thus $v_1 = (1,0)$ & $v_2 = (0,1)$ $T(v_1) = T(1,0) = (2(1) + 0, 1)$ [Using given definition of T] $= (2, 1)$ $= 2(1,0) + 1(0,1)$ $= 2v_1 + 1v_2$ Thus $T(v_1) = 2v_1 + 1v_2$ ---(1) $T(v_2) = T(0,1) = (2(0) + 1, 0)$ [Using given definition of T] $= (1, 0)$ $= 1(1,0) + 0(0,1)$ $= 1v_1 + 0v_2$ Thus $T(v_2) = 1v_1 + 0v_2$ ---(2) From (1) and (2), the required matrix of linear transformation T relative to α is given by $[T]_\alpha = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$	2	K1	CO4
7.	If T is a linear operator on $P_2(\mathbb{R})$ defined by $T(f(x)) = f(x) + f(2)x$, write the matrix of T relative to the standard basis of $P_2(\mathbb{R})$. Here the linear operator T on $V = P_3(\mathbb{R})$, $T(f(x)) = f(x) + f(2)x$. The standard basis of the vector space $P_3(\mathbb{R})$ is given by $\alpha = \{1, x, x^2\}$. $T(1) = 1 + 1 \cdot x = 1.1 + 1 \cdot x + 0 \cdot x^2$ $T(x) = x + 2 \cdot x = 3x = 0.1 + 3 \cdot x + 0 \cdot x^2$ $T(x^2) = x^2 + 4x = 0.1 + 4 \cdot x + 1 \cdot x^2$ $\Rightarrow T \begin{bmatrix} 1 \\ x \\ x^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ x^2 \end{bmatrix}$ From the above, we have required matrix of linear transformation T relative to α is given by $[T]_\alpha = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$	2	K1	CO4
8.	Define eigen values and eigen vectors of a linear transformation. Let T be a linear operator defined on a vector space V. A non-zero vector $x \in V$ is called an eigenvector of T, if there exists a scalar λ called eigenvalue such that $T(x) = \lambda x$.	2	K2	CO4
9.	Define characteristic equation. Let T be a linear operator defined on an n-dimensional vector space V	2	K2	CO4



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	with ordered basis β . The characteristic polynomial of $A = [T]_\beta$ is given by $ A - \lambda I $ and the characteristic equation is given by $ A - \lambda I = 0$			
10.	Find the eigenvalues of the linear operator T on $\mathbb{R}^2$ by $T(a, b) = (-2a + 3b, -10a + 9b)$. By Standard basis of $\mathbb{R}^2$, Matrix of T is given by $[T]_\alpha = \begin{bmatrix} -2 & 3 \\ -10 & 9 \end{bmatrix}$. Now Characteristic equation of T is $ T - \lambda I = 0$ $\Rightarrow \lambda^2 - 7\lambda + 12 = 0$ $\Rightarrow (\lambda - 3)(\lambda - 4) = 0$ $\Rightarrow \lambda = 3, 4$ are the required eigenvalues of T	2	K2	CO4
11.	Prove that a scalar λ is an eigenvalue of Linear operator T on V if and only if $ T - \lambda I = 0$. Proof: Given that λ is an eigenvalue of T $\Leftrightarrow T(v) = \lambda v, v \neq 0$ Since by defn. of eigenvalue of T $\Leftrightarrow T(v) - \lambda v = 0,$ $\Leftrightarrow (T - \lambda I)(v) = 0,$ $\Leftrightarrow v \in N(T - \lambda I), v \neq 0, \text{ (Since by defn of } N(T))$ $\Leftrightarrow N(T - \lambda I) \neq \{0\}$ $\Leftrightarrow T - \lambda I$ is not one to one, Since by known Theorem $\Leftrightarrow T - \lambda I$ is not invertible $\Leftrightarrow T - \lambda I = 0$.	2	K1	CO4
12.	Define diagonalizability of a linear transformation. A linear operator T on a finite dimensional vector space V is diagonalizable if and only if there exists an ordered basis β for V consisting of eigenvectors of T .	2	K1	CO4
13.	Write the test for diagonalizability. A given square matrix is diagonalizable if its eigenvalues are unique and non-repeating (i.e.) all eigenvalues should have multiplicity 1.	2	K1	CO4
14.	Is the linear operator $T(a, b) = (4a + b, -a + 2b)$ on $\mathbb{R}^2$ diagonalizable? By Standard basis of $\mathbb{R}^2$, Matrix of T is given by $[T]_\alpha = \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}$. Now Characteristic equation of T is $ T - \lambda I = 0$ $\Rightarrow \lambda^2 - 6\lambda + 9 = 0$ $\Rightarrow (\lambda - 3)(\lambda - 3) = 0$ $\Rightarrow \lambda = 3, 3$ are the required eigenvalues of T Here the multiplicity of 3 is 2 $\Rightarrow T$ is not diagonalizable.	2	K2	CO4
15.	Is the linear operator $T(a, b) = (3a, -a + 2b)$ on $\mathbb{R}^2$ diagonalizable? By Standard basis of $\mathbb{R}^2$, Matrix of T is given by $[T]_\alpha = \begin{bmatrix} 3 & 0 \\ -1 & 2 \end{bmatrix}$. Now Characteristic equation of T is $ T - \lambda I = 0$	2	K2	CO4



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	$\Rightarrow \lambda^2 - 5\lambda + 6 = 0$ $\Rightarrow (\lambda - 2)(\lambda - 3) = 0$ $\Rightarrow \lambda = 2, 3$ are the required eigenvalues of T Here the multiplicity of 2 is 1 and multiplicity of 3 is 1 $\Rightarrow T$ is diagonalizable.			
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Part B

S. No	Question	Marks	BTL	CO
1.	(i) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation defined by $T(a_1, a_2) = (2a_1 - a_2, 3a_1 + 4a_2, a_1)$, find $[T]_{\alpha}^{\beta}$ given that α & β are the standard basis of $\mathbb{R}^2$ & $\mathbb{R}^3$ respectively. (ii) If $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is a linear transformation defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$, find matrix of T relative to the standard basis of $\mathbb{R}^3$ & $\mathbb{R}^2$ respectively. (iii) If $T: M_{2 \times 2}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ is a linear transformation defined by $T(A) = A^t$, find matrix of T relative to the standard basis of $M_{2 \times 2}(\mathbb{R})$. (iv) If $T: P_2(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ is a linear transformation defined by $T(f(x)) = \begin{bmatrix} f'(0) & 2f(1) \\ 0 & f''(3) \end{bmatrix}$, find matrix of T relative to the standard basis of $P_2(\mathbb{R})$ & $M_{2 \times 2}(\mathbb{R})$. (v) If $T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ is a linear transformation defined as $T(f(x)) = xf(x) + f'(x)$ find the matrix of T relative to the standard basis of $P_2(\mathbb{R})$ & $P_3(\mathbb{R})$ (vi) If $T: M_{2 \times 2}(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ is a linear transformation defined as $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a + b) + (2d)x + bx^2$ find $[T]_{\alpha}^{\beta}$ where α & β are standard basis of $M_{2 \times 2}(\mathbb{R})$ & $P_2(\mathbb{R})$ respectively.	8	K4	CO4
2.	(i) Find the matrix of the linear transformation T on $\mathbb{R}^3$ defined by $T(a, b, c) = (2b + c, a - 4b, 3a)$ with respect to the ordered basis $\beta = \{(1,1,1), (1,1,0), (1,0,0)\}$ (ii) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation defined by $T(a_1, a_2) = (a_1 - a_2, a_1, 2a_1 + a_2)$. Let β be the standard ordered basis for $\mathbb{R}^2$ and $\gamma = \{(1,1,0), (0,1,1), (2,2,3)\}$ be ordered basis of $\mathbb{R}^3$. Compute $[T]_{\gamma}^{\beta}$. (iii) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation defined by $T(a_1, a_2, a_3) = (2a_1 + 3a_2 - a_3, a_1 + a_3)$. If $\alpha = \{(1,1,0),$	8	K4	CO4



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	$(0,1,1), (2,2,3)\}$, $\beta = \{(1,2), (2,3)\}$ are ordered basis of $\mathbb{R}^3, \mathbb{R}^2$ respectively, then compute $[T]_{\alpha}^{\beta}$.			
3.	(i) If T is linear on $V = \mathbb{R}^2$ defined by $T(a, b) = (-2a + 3b, -10a + 9b)$ find the eigen values of T and an ordered basis β for V such that $[T]_{\beta}$ is a diagonal matrix. (ii) If T is linear on $V = \mathbb{R}^3$ defined by $T(a, b, c) = (7a - 4b + 10c, 4a - 3b + 8c, -2a + b - 2c)$ find the eigen values of T and an ordered basis β for V such that $[T]_{\beta}$ is a diagonal matrix. (iii) If T is linear on $V = \mathbb{R}^3$ defined by $T(a, b, c) = (-4a + 3b - 6c, 6a - 7b + 12c, 6a - 6b + 11c)$ find the eigen values of T and an ordered basis β for V such that $[T]_{\beta}$ is a diagonal matrix.	16	K4	CO4
4.	(i) Let T on $\mathbb{R}^2$ defined as $T(a, b) = (a + 4b, 3a + 2b)$ be a linear transformation. Check for diagonalizability of T and find basis β such that $[T]_{\beta}$ is a diagonal matrix. If diagonalizable, find an invertible matrix Q and diagonal matrix D. (ii). Let T on $\mathbb{R}^2$ defined as $T(a, b) = (5a + 4b, a + 2b)$ be a linear transformation. Check for diagonalizability of T and find basis β such that $[T]_{\beta}$ is a diagonal matrix. If diagonalizable, find an invertible matrix Q and diagonal matrix D.	16	K4	CO4



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DEPARTMENT OF SCIENCE AND HUMANITIES (MATHEMATICS DIVISION)

QUESTION BANK

Regulation	:	2023
Course Name	:	APPLIED LINEAR ALGEBRA
Course Code	:	JMA2402
Year / Semester / Branch		II/ IV/ CSE/IT/AI&ML/AI&DS/CS&BS/CS/ECE
Unit No	:	V
Unit Title	:	INNER PRODUCT SPACES
No. of Hours	:	12
Weightage in Exam (%)		20 %

Part A

S. No	Question	Marks	BTL	CO
1.	Define inner product space. Let V be a vector space with field F, which is either field of real numbers or field of complex numbers. The vector space V is said to be inner product space, the operation inner product can be defined satisfying the following laws, for all $x, y, z \in V$ & $c \in F$ (i) $(x + z, y) = (x, y) + (z, y)$ (ii) $(cx, y) = c(x, y)$ (iii) $\overline{\langle x, y \rangle} = \langle y, x \rangle$ (iv) $(x, x) > 0$ if $x \neq 0$	2	K1	CO5
2.	Show that $(x, y + z) = (x, y) + (x, z)$ given $x, y, z \in V$ an inner product space defined over field F. Consider $\begin{aligned} \langle x, y + z \rangle &= \overline{\langle y + z, x \rangle} \\ &= \overline{\langle y, x \rangle + \langle z, x \rangle} \\ &= \overline{\langle y, x \rangle} + \overline{\langle z, x \rangle} \\ &= (x, y) + (x, z) \end{aligned}$ Thus $(x, y + z) = (x, y) + (x, z)$	2	K2	CO5
3.	Define norm of a vector. Let V be an inner product space. For the vector $x \in V$ norm or length is defined as the positive square root of (x, x) (i.e.) $\ x\ = \sqrt{(x, x)}$	2	K1	CO5



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4.	State and prove any two properties of norm. Let V be an inner product space. Then for all $x, y \in V$ & $c \in F$ we have the following properties (i) $\ cx\ = c \ x\$ Consider $\ cx\ ^2 = (cx, cx)$ $= c\bar{c}(x, x)$ [By linearity] $= c ^2\ x\ ^2$ [Since $c\bar{c} = c ^2$ & $(x, x) = \ x\ ^2$] Thus taking square root we get $\ cx\ = c \ x\$ (ii) $\ x\ = 0$ if and only if $x = 0$ Consider $\ x\ = 0 \Leftrightarrow \ x\ ^2 = 0$ $\Leftrightarrow (x, x) = 0$ $\Leftrightarrow x = 0$ Thus $\ x\ = 0$ if and only if $x = 0$	2	K2	CO5
5.	Define distance between two vectors. Let V be an inner product space. Then the distance between two vectors x and y is defined as $d(x, y) = \ x - y\ = (x - y, x - y)$	2	K1	CO5
6.	State Cauchy-Schwarz inequality. In an inner product space $V(F)$, then $\langle x, y \rangle \leq \ x\ \ y\$, for $x, y \in V$	2	K1	CO5
7.	State Triangle inequality and Parallelogram law. In an inner product space $V(F)$, then (i) Triangle inequality $\ x + y\ \leq \ x\ + \ y\$, for $x, y \in V$ (ii) Parallelogram law $\ x + y\ ^2 + \ x - y\ ^2 = 2\ x\ ^2 + 2\ y\ ^2$, for $x, y \in V$	2	K1	CO5
8.	State Polar identities over the real and complex fields In an inner product space $V(F)$, then $\forall x, y \in V$ (i) Polar identities over the real field $4 \langle x, y \rangle = \ x + y\ ^2 - \ x - y\ ^2$, (ii) Polar identities over the complex field $4 \langle x, y \rangle = \{(\ x + y\ ^2 - \ x - y\ ^2) + i(\ x + iy\ ^2 - \ x - iy\ ^2)\}$	2	K1	CO5
9.	State Pythagorean theorem. Let V be a real inner product space & $x, y \in V$, then x & y are orthogonal if and only if $\ x + y\ ^2 = \ x\ ^2 + \ y\ ^2$	2	K1	CO5
10.	Show that $d(x, y) \leq d(x, z) + d(z, y)$ given $x, y, z \in V$, an inner product space defined over the field F. Consider $d(x, y) = \ x - y\$ $= \ (x - z) + (z - y)\$ $\leq \ x - z\ + \ z - y\$ $= d(x, z) + d(z, y)$ Thus $d(x, y) \leq d(x, z) + d(z, y)$	2	K2	CO5



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11.	Define orthogonal vectors. Let V be an inner product space and $x, y \in V$. Then x is said to be orthogonal to y if $\langle x, y \rangle = 0$	2	K1	CO5
12.	Show that orthogonality of vectors is a symmetric relation. To prove: x is orthogonal to $y \Rightarrow y$ is orthogonal to x Let x be orthogonal to y (i.e.) $\langle x, y \rangle = 0$. Consider $\overline{\langle x, y \rangle} = \overline{0}$ $\Rightarrow \overline{\langle x, y \rangle} = \overline{0}$ $\Rightarrow \langle y, x \rangle = 0$ [Since $\overline{\langle x, y \rangle} = \langle y, x \rangle$ & $\overline{0} = 0$] $\Rightarrow y$ is orthogonal to x . Thus x is orthogonal to $y \Rightarrow y$ is orthogonal to x	2	K2	CO5
13.	Define an orthogonal set. A subset S of an inner product space V is said to be orthogonal set if any two distinct vectors in S are orthogonal (i.e.) for $x, y \in S$ if $x \neq y$ then $\langle x, y \rangle = 0$	2	K1	CO5
14.	Define an orthonormal set. A subset S of an inner product space V is said to be orthonormal set if (i) For all $x \in S$, $\ x\ = 1$ (i.e.) $\langle x, x \rangle = 1$ For $x, y \in S$ if $x \neq y$ then $\langle x, y \rangle = 0$	2	K1	CO5
15.	Write the algorithm of Gram-Schmidt process to find an orthonormal basis $\{v_1, v_2, v_3\}$ using the set $\{u_1, u_2, u_3\}$. Step1: $v_1 = \frac{u_1}{\ u_1\ }$ Step2: $v_2 = \frac{w_2}{\ w_2\ }$, where $w_2 = u_2 - \langle u_2, v_1 \rangle v_1$ Step3: $v_3 = \frac{w_3}{\ w_3\ }$, where $w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$	2	K2	CO5

Part B

s.n	Question	Marks	BTL	CO
1.	(i) Show that $F^n(\mathbb{C})$ is an inner product space with respect to the inner product defined as $\langle x, y \rangle = \sum^n x_i \bar{y}_i$ where $x = (x_1, x_2, \dots, x_n)$ & $y = (y_1, y_2, \dots, y_n)$ (ii) Show that $F^n(\mathbb{R})$ is an inner product space with respect to the inner product defined as $\langle x, y \rangle = \sum^n x_i y_i$ where $x = (x_1, x_2, \dots, x_n)$ & $y = (y_1, y_2, \dots, y_n)$ (iii) Prove that $\langle f, g \rangle = \int_0^1 f(x) \overline{g(x)} dx$ is an inner product where $f, g \in \mathbb{C}[0,1]$ a vector space of complex valued continuous functions on $[0,1]$. (iv) Prove that $\langle f, g \rangle = \int_0^1 f(x) g(x) dx$ is an inner product where $f, g \in \mathbb{R}[0,1]$ a vector space of real valued continuous functions on $[0,1]$	8	K4	CO5



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2.	(i) State and prove Cauchy-Schwartz inequality. (ii) State and prove triangle inequality and parallelogram law on an inner product space V. (iii) State and prove polar identity on a real inner product space V. (iv) State and prove Pythagorean theorem on a real inner product space V.	8	K4	CO5
3.	(i) Apply Gram-Schmidt orthogonalization process to the subset $S = \{(1,1,1), (0,1,1), (0,0,1)\}$ of $\mathbb{R}^3$ to obtain an orthonormal basis. Also find Fourier coefficients of $(1,0,1)$ relative to this orthonormal basis. (ii) Apply Gram-Schmidt orthogonalization process to the subset $S = \{(1,0,1), (0,1,1), (1,3,3)\}$ of $\mathbb{R}^3$ to obtain an orthonormal basis. Also find Fourier coefficients of $(1,1,2)$ relative to this orthonormal basis. (iii) Apply Gram-Schmidt process to obtain an orthonormal basis for $V = P_2(\mathbb{R})$ with inner product defined by $\langle f, g \rangle = \int_0^1 f(x)g(x)dx.$ Also find Fourier coefficients of $h(x) = 1 + x$ relative to this orthonormal basis. (iv) Apply Gram-Schmidt process to the subset $S = \left\{ \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 9 \\ 5 & -1 \end{bmatrix}, \begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix} \right\}$ of $M_{2 \times 2}(\mathbb{R})$ to obtain orthonormal basis using the Frobenius inner product. Also find the Fourier coefficients of $\begin{bmatrix} -1 & 27 \\ -4 & 8 \end{bmatrix}$ relative to this orthonormal basis.	16	K4	CO5